

CAR RACING CHAMPIONSHIP

Product Requirements Document (PRD)

1. Project Overview

Create a browser-based Car Racing Game that delivers a fast-paced and engaging racing experience. Players control a racing vehicle, avoid obstacles, collect rewards, and compete for the highest score.

The game should feature smooth controls, realistic vehicle movement, attractive visuals, responsive gameplay, and support for both desktop and mobile devices.

Players start the game using a Start Race button and may end the session using an End Race button at any time. No login or registration is required.

2. Game Specifications

Feature	Requirement
Game Type	Browser-based Racing Game
Players	Single Player
Platform	Modern Web Browsers
Controls	Keyboard and Touch Controls
Camera View	Top-Down Racing View
Game Mode	Endless Racing
Difficulty	Progressive
Audio Control	Mute / Unmute
Device Support	Desktop and Mobile

3. Gameplay Rules

3.1 Objective

- Drive as far as possible.
- Avoid collisions.
- Collect coins and power-ups.
- Achieve the highest score.

3.2 Player Vehicle

- Controlled by the player.
- Can move left and right.
- Speed gradually increases.

3.3 Obstacles

- Traffic Cars
- Road Barriers
- Oil Spills
- Construction Blocks

3.4 Power-Ups

- Speed Boost
- Shield Protection
- Coin Multiplier
- Score Bonus

3.5 Game Start

1. User opens the website.
2. Main Menu appears.
3. User clicks Start Race.
4. Countdown animation begins.
5. Race starts.

3.6 Game End

The game ends when:

- Vehicle crashes into an obstacle.
- Vehicle crashes into traffic.
- Health reaches zero.
- User clicks the End Race button.

After game completion:

- Final score is displayed.
 - Distance travelled is displayed.
 - Coins collected are displayed.
 - High score is displayed.
 - Restart Race button appears.
 - Return to Main Menu button appears.
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4. Functional Requirements

FR-1 Start Game System

- Start Race button available.
- Clicking Start Race begins gameplay.
- Countdown animation displayed.
- No login required.

FR-2 Vehicle Control System

- Move Left
- Move Right
- Smooth movement
- Responsive controls

FR-3 Traffic System

- AI traffic vehicles.
- Random spawning.
- Increasing traffic density.

FR-4 Obstacle System

- Random obstacle generation.
- Dynamic difficulty scaling.

FR-5 Coin Collection System

- Coins spawn randomly.
- Coin count displayed.
- Coins increase score.

FR-6 Power-Up System

- Random spawning.
- Temporary effects.
- Visual indicators.

FR-7 Score System

- Distance score.
- Coin bonus score.
- Survival time bonus.

FR-8 High Score System

- Save highest score.
- Display current score.
- Display final score.

FR-9 Audio System

- Background music.
- Engine sound.
- Coin collection sound.
- Crash sound.
- Mute button.
- Unmute button.

FR-10 Responsive Support

- Desktop compatibility.
- Mobile compatibility.
- Touch support.
- Adaptive layouts.

FR-11 End Game System

- End Race button available.
 - Clicking End Race immediately stops gameplay.
 - Final score screen displayed.
 - Restart Race button available.
 - Return to Main Menu button available.
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5. User Interface Requirements

Main Menu

- Game Logo
- Start Race Button
- Audio Controls
- Instructions Button

Gameplay Screen

Top Section

- Current Score
- Distance Travelled
- Coin Counter

Center Section

- Racing Track
- Player Vehicle
- Obstacles
- Traffic Vehicles

Bottom Section

- Control Instructions
- Status Messages

Additional Controls

- End Race Button
- Mute Button
- Unmute Button

Game Over Screen

- Final Score
 - Distance Travelled
 - Coins Collected
 - High Score
 - Restart Race Button
 - Return to Main Menu Button
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6. Visual Requirements

Environment

- Highway Track
- Road Markings
- Moving Background
- Dynamic Weather Effects

Vehicle Graphics

- Player Racing Car
- Traffic Vehicles
- Power-Up Effects

UI Design

- Modern Racing Theme
- Smooth Animations
- Premium Graphics

- Racing Dashboard Style
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7. Animation Requirements

Car Movement Animation

- Smooth lane switching.
- Responsive steering effects.
- Vehicle tilt animation.

Collision Animation

- Impact effects.
- Vehicle shake effect.
- Crash particles.

Reward Animation

- Coin collection effect.
- Power-up activation effect.
- Score increase effect.

Menu Animation

- Button hover effects.
 - Page transitions.
 - Countdown animation.
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8. Sound Requirements

Required Sounds

- Engine Sound
- Coin Collection Sound

- Power-Up Sound
- Crash Sound
- Menu Click Sound
- Game Over Sound

Audio Controls

- Mute Button
 - Unmute Button
 - Audio Status Indicator
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9. Accessibility Requirements

- Large buttons.
 - Clear text.
 - High contrast visuals.
 - Mobile-friendly controls.
 - Audio indicators.
 - Touch-friendly interface.
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10. Acceptance Criteria

#	Requirement
1	Car Racing Game is playable
2	Start Race button launches the game
3	End Race button ends the game
4	Desktop devices are supported
5	Mobile devices are supported
6	Player can control the vehicle
7	Obstacles spawn correctly
8	Traffic vehicles function correctly

- 9 Coins can be collected
 - 10 Power-ups function correctly
 - 11 Score updates properly
 - 12 Mute/Unmute functionality works
 - 13 Game Over screen displays final statistics
 - 14 Restart Race button functions correctly
 - 15 Return to Main Menu button functions correctly
 - 16 Responsive design functions correctly
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11. Deliverables

#	Item	Format	Notes
1	Complete Project Source Code	ZIP	Full game source code
2	Playable Web Build	Folder	Browser-ready version
3	README	MD	Setup and gameplay instructions
4	Vehicle Controller Module	JS	Vehicle movement system
5	Traffic System Module	JS	AI traffic logic
6	Score System Module	JS	Scoring functionality
7	Audio System Module	JS	Audio controls
8	UI Assets	Folder	Images, icons, graphics
9	Sound Assets	Folder	Audio files
10	Asset Credits	TXT	Asset attribution

12. Folder Structure

Car_Racing_Championship/

|— README.md

|— Dockerfile

|— index.html

|— css/

| |— styles.css

|— js/

| |— game.js

| |— vehicle.js

| |— traffic.js

| |— scoring.js

| |— audio.js

| |— ui.js

|— assets/

| |— images/

| |— sounds/

| |— vehicles/

| |— animations/

|— docs/

| |— asset_credits.txt

|— build/

|— playable_web_version/

13. Final Deliverables Checklist

Category	Description	Files
Source Code	Complete game source code	1
Web Build	Playable browser version	1
Documentation	README and setup guide	1
Assets	Images, sounds, animations	Multiple
Audio System	Mute and Unmute functionality	1
Responsive Design	Desktop and mobile support	1
Credits	Asset attribution file	1
Final Submission	Complete project package	1 ZIP

Final Acceptance Checklist

- Start Race button works.
- End Race button works.
- Desktop support works.
- Mobile support works.
- Audio controls function correctly.
- Scoring system functions correctly.
- Responsive design is implemented.
- Final score screen is displayed.
- Deliverables are complete.
- Project package is ready for submission.